

Database Assignment 9

Q1) Create an index that will enable a user to pull orders for '1990-10-03' out of the Orders table quickly.

```
w1_99099_vaibhavi>CREATE INDEX idx_orders_odate on orders(odate);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0

w1_99099_vaibhavi>select odate from orders where odate= '1990-10-03';
+-----+
| odate |
+-----+
| 1990-10-03 |
| 1990-10-03 |
| 1990-10-03 |
| 1990-10-03 |
| 1990-10-03 |
+-----+
5 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q2) If the Order table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
w1_99099_vaibhavi>ALTER TABLE Orders
-> ADD CONSTraint uq_orders_onum UNIQUE(onum);
ERROR 1061 (42000): Duplicate key name 'uq_orders_onum'
w1_99099_vaibhavi>SELECT * FROM ORDERS;
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

```
10 rows in set (0.00 sec)

w1_99099_vaibhavi>SELECT * FROM orders WHERE odate = '1990-10-03';
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007

```
5 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q3) Create an index that would permit salesperson to retrieve his orders

```
w1_99099_vaibhavi>CREATE INDEX idx1_orders_snum
-> ON Orders (snum);
Query OK, 0 rows affected, 1 warning (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 1

w1_99099_vaibhavi>SELECT * FROM orders WHERE snum = 1001;
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

```
3 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q4) Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
w1_99099_vaibhavi>CREATE INDEX indx_customer_sum_rating ON customers(snum, rating);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Q5) Create an index to speed up searching order on a given date by given customer.

```
w1_99099_vaibhavi>SELECT *
-> FROM ORDERS
-> WHERE odate = '1990-10-04'
-> AND
-> cnum = 2004;
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

w1_99099_vaibhavi>
```